CHART THEORY 17.
On a Lambert conformal conic chart, the distance between parallels of latitude spaced the same number of degrees apart?

a. Increases between and reduces outside of the standard parallels.
b. Reduces between, and expands outside, the standard parallels.
c. Is constant between the standard parallels.
d. Is constant outside of the standard parallels.

CHART THEORY 18.
Which one of the following statements is correct concerning the appearance of great circles, with the exception of meridians, on a Polar Stereographic chart whose tangency is at the pole?

a. The higher the latitude the closer they approximate to a straight line.
b. The lower the latitude the closer they approximate to straight lines.
c. Never approximate to straight lines.
d. Always approximate to straight lines.

CHART THEORY 19.
Which one of the following describes the appearance of rhumb lines, except meridians, on a Polar Stereographic chart?

a. Straight lines.
b. Curves concave to the Pole.
c. Approximately straight lines.
d. Curves closer to the pole than the equivalent Great Circle.

CHART THEORY 20.
What is the value of the convergence factor on a Polar Stereographic chart?

a. 2.0.
b. 1.5.
c. 1.0.
d. 0.5.

CHART THEORY 21.
On a Direct Mercator, rhumb lines are?

a. Straight lines.
b. Cuves concave to the nearest pole.
c. Curves convex to the nearest pole.
d. Curves concave to the Equator.

CHART THEORY 22.
A perfectly straight line on a Lamberts chart is?

a. A parallel of latitude.
b. A rhumb line.
c. A meridian of longitude.
d. A great circle.

CHART THEORY 23.
Which of the following statements best describes how scale varies on a Mercator chart?

a. It is correct on the standard parallels, but expands outside them, and contracts within them.
b. Expands as the secant of the East/West great circle distance.
c. Expands as the secant (½ x co-latitude).
d. Expands directly with the secant of the latitude.

CHART THEORY 24.
How do rhumb lines, (other than meridians), appear on Polar Stereographic charts?

a. As straight lines.
b. As lines concave to the nearer pole.
c. As lines convex to the nearer pole.
d. As ellipses around the pole.

CHART THEORY 25.
Where is the convergency correct on a Transverse Mercator chart?

a. At the datum meridian and the Equator.
b. Only at the Equator and Poles.
c. Only at the datum meridian.
d. At the Parallel of Origin.

CHART THEORY 26.
In which of the following projections will a plane surface touch the reduced earth at one of the Poles?

c. Lambert's.
d. Direct Mercator.
a. Transverse Mercator.
d. Stereographic.

CHART THEORY 27.
How are great circles shown on a direct Mercator chart?

a. Curves concave to the nearest pole.
b. Curves convex to the nearest pole.
c. Straight lines.
d. Rhumb lines.

CHART THEORY 28.
The scale on a Lambert's conformal conic chart?

a. Is constant.
b. Is constant along a parallel of latitude.
c. Is constant along a meridian of longitude.
d. Varies slightly as a function of longitude and latitude.

CHART THEORY 29.
The scale is correct on a Transverse Mercator chart?

a. Along the datum meridian and meridians at 90° to it.
b. At the prime meridian.
c. Along the great circle of tangency.
d. At the poles and the equator.

CHART THEORY 30.
What is the symbol for an FIR boundary?

a. .....
b. -
c. - . - . - .

a. 1.
b. 3.
c. 7.
d. 14.

CHART THEORY 38.
Please refer to 58. Which of the following is the symbol for an exceptionally high (over 300m above terrain) lighted obstruction?

a. 9.
b. 11.
c. 10.
d. 15.

CHART THEORY 39.
Please refer to page 58. Which of the following is the symbol for an unlighted obstacle?

a. 9.
b. 11.
c. 10.
d. 15.

CHART THEORY 40.
Please refer to page 58. Which of the following is the symbol for a group of obstacles?

a. 9.
b. 11.
c. 10.
d. 15.

CHART THEORY 41.
Please refer to page 58. Which of the following is the symbol for a VOR DME?

a. 4.
b. 5.
c. 7.
d. 14.

CHART THEORY 42.
Please refer to page 58. Which of the following is the symbol for a VOR TAC?

a. 4.
b. 5.

c. 7.
d. 14.

CHART THEORY 43.
Please refer to page 58. Which of the following is the symbol for a TACAN?

a. 4.
b. 5.
c. 7.
d. 14.

CHART THEORY 44.
Please refer to page 58. Which of the following is the symbol for a DME?

a. 4.
b. 5.
c. 7.
d. 14.

CHART THEORY 45.
Please refer to page 58. Which of the following is the symbol for an NDB?

a. 1.
b. 2.
c. 3.
d. 4.

CHART THEORY 46.
Please refer to page 58. Which of the following is the symbol for a lighted group of obstacles?

a. 9.
b. 10.
c. 11.
d. 12.

CHART THEORY 47.
Please refer to page 58. Which of the following is the symbol for a lighted Obstacle?

a. 9.
b. 10.
c. 11.
d. 12.

CHART THEORY 48.
Please refer to page 58. Which of the following is the symbol for a lighted group of obstacles?

a. 6.
b. 10.
c. 11.
d. 12.

CHART THEORY 49.
The nominal scale of a Lambert conformal conic chart is the?

a. Scale at the standard parallels.
b. Scale at the parallel of origin.
c. Scale at the poles.
d. Constant at all points on the chart.

CHART THEORY 50.
What type of projection is generally used for charts used for navigation in North and South polar areas?

a. Lanbert's Conformal.
b. Polar Stereographical projection.
c. Direct Mercator.
d. Oblique Mercator.

CIRCLES 1.
A great circle track joins position A (59° S 141°W) and B (61° S 148°W). What is the difference between the great circle track at A and B?

a. It increases by 6°.
b. It decreases by 6°.
c. It increases by 3°.
d. It decreases by 3°.

CIRCLES 2.
Given: Position A 45° N, ?° E, Position B 45° N, 45° 15'E Distance A-B = 280 NM, B is to the East of A, What is the longitude of position A?

a. 38° 39'E.
b. 49° 57'E.
c. 51° 51'E.
d. 40° 33'E.

CIRCLES 3.
A Lambert conformal conic chart has a constant of the cone of 0.75. The initial course of a straight-line track drawn on this chart from A (40° N 050° W) to B is 043° (T) at A; course at B is 055° (T). What is the longitude of B?

a. 34°W.
b. 36°W.
c. 38°W.
d. 41°W.

CIRCLES 4.
What is the great circle track (measured from the starting position) from 70°S 030°W, to 70°S 060°E?

a. 048°T.
b. 090°T.
c. 132°T.
d. 228°T.

CIRCLES 5.
What is the longitude of a position 6 NM to the east of 58° 42'N 094° 00'W?

a. 093° 53.1'W.
b. 094° 12.0'W.
c. 093° 48.5'W.
d. 093° 54.0'W.

CIRCLES 6.
As the INS position of the departure aerodrome, coordinates 35° 32.7'N 139° 46.3'W are input instead of 35° 32.7'N 139° 46.3'E. When the aircraft subsequently passes point 52° N 180°W, the longitude value shown on the INS will be?

a. 080° 27.4'W.
b. 099° 32.6'W.
c. 099° 32.6'E.
d. 080° 27.4'E.

CIRCLES 7.
An aircraft flies the following rhumb line tracks and distances from position 04° 00'N 030° 00'W: 600 NM South, then 600 NM East, then 600 NM North, then 600 NM West. The final position of the aircraft is?

a. 04° 00'N 030° 00'W.
b. 03° 58'N 030° 02'W.

c. 04° 00'N 029° 55'W.
d. 04° 00'N 030° 02'W.

CIRCLES 8.
Which of the following differences in latitude will produce the greatest difference between the initial great circle track and the mean great circle track, between two points, which are separated by 10° change of longitude?

a. 30°S and 25°S.
b. 30°S and 30°N.
c. 60°N and 55°N.
d. 60°N and 60°S.

CIRCLES 9.
What is the final position after the following rhumb line tracks and distances have been followed from position 60° 00'N 030° 00'W? South for 3600 NM, East for 3600 NM, North for 3600 NM, West for 3600 NM?

a. 59° 00'N 090° 00'W.
b. 60° 00'N 090° 00'W.
c. 60° 00'N 030° 00'E.
d. 59° 00'N 060° 00'W.

CIRCLES 10.
The Great Circle bearing of 'B' (70° S 060° E), from 'A' (70° S 030° W), is approximately?

a. 150° (T).
b. 090° (T).
c. 315° (T).
d. 135° (T).

CIRCLES 11.
An aircraft at position 60° N 005° W tracks 090° (T) for 315 km. On completion of the flight the longitude will be?

a. 002° 10'W.
b. 000° 15'E.
c. 000° 40'E.
d. 005° 15'E.

CIRCLES 12.
An aircraft autopilot is coupled to the INS steering from Waypoint 1 at 60°N 040°W. to Waypoint 2 at 60°N 030°W. What is the approximate latitude of the aircraft on passing 35°W?

a. 60°05'N.
b. 60°19'N.
c. 60°30'N.
d. 59°15'N.

CIRCLES 13.
A Lambert conformal conic chart has a constant of the cone of 0.80. A straight line course drawn on this chart from A (53° N 004°W) to B is 080° at A Course at B is 092° (T). What is the longitude of B?

a. 011° E.
b. 009° 36'E.
c. 008° E.
d. 019° E.

CIRCLES 14.
An aircraft is flying with the aid of an inertial navigation system (INS) connected to the autopilot. The following points have been entered in the INS computer: WPT 1: 60° N 030°W, WPT 2: 60° N 020°W. When 025° W has been passed, the latitude shown on the display unit of the inertial navigation system will be?

a. 60° 10.0'N.
b. 59° 49.0'N.
c. 60° 11.0'N.
d. 60° 05.0'N.

CIRCLES 15.
The 'departure' between positions 60°N 160° E and 60° N 'x' is 900 NM. What is the longitude of 'x'?

a. 170°W.
b. 140°W.
c. 145° E.
d. 175° E.

CIRCLES 16.
Given: The autopilot is coupled to the Inertial Navigation System steering from Waypoint 1 at 60°N 030°W, to Waypoint 2 at 60°N 020°W, to Waypoint 3 at 60°N 010°W. What is the approximate track change on passing WPT 2?

a. Zero.
b. 4° decrease.
c. 9° decrease.
d. 9° increase.

CIRCLES 17.
An aircraft autopilot is coupled to the INS steering from Waypoint 1 at 60°N 050°W, to Waypoint 2 at 60°N 040°W. What is the approximate latitude of the aircraft on passing 45°W?

a. 60°30'N.
b. 59°15'N.
c. 60°05'N.
d. 60°19'N.

CIRCLES 18.
The 'departure' between positions 60°N 160° E and 60° N 'x' is 1200 NM. What is the longitude of 'x'?

a. 160°W.
b. 140°W.
c. 145° E.
d. 175° E.

CIRCLES 19.
The 'departure' between positions 60°N 60° E and 60° N 'x' is 1800 NM East. What is the longitude of 'x'?

a. 145° E.
b. 145° E.
c. 120°E.
d. 120°W.

CIRCLES 20.
An aircraft with its autopilot connected to its inertial system, flies from point A at 45°S 010°W, to point B at 45°S 030°W. What is the true course of the aircraft (to the nearest degree) when it arrives at B?

a. 265°.
b. 275°.
c. 277°.
d. 286°.

CIRCLES 21.
The automatic flight control system (AFCS) in an aircraft is coupled to the guidance outputs from an inertial navigation system (INS). The aircraft is flying between inserted waypoints No. 3 (55° 00'N 020° 00'W) and No.4 (55° 00'N 030° 00'W). With DSRTK/STS selected on the CDU, to the nearest whole degree, the initial track read-out from waypoint No. 3 will be?

a. 266°.
b. 270°.
c. 274°.
d. 278°.

CIRCLES 22.
An aeroplane flies from A (59° S 142° W) to B (61° S 148° W) with a TAS of 480 kt. The autopilot is engaged and coupled with an Inertial Navigation System in which AB track is active. On route AB, the true track?

a. Varies by 10°.
b. Decreases by 6°.
c. Varies by 4°.
d. Increases by 5°.

CIRCLES 23.
The automatic flight control system is coupled to the guidance outputs from an inertial navigation system. Which pair of latitudes will give the greatest difference between initial track read-out and the average true course given, in each case, a difference of longitude of 10°?

a. 60° N to 60°N.
b. 60° N to 50°N.
c. 30° S to 30°N.
d. 30° S to 25°S.

CIRCLES 24.
Position A is 55°N 030°W and B is 54°N 020°W. What is the rhumb line bearing from A to B, if the great circle track from A to B, measured at A, is 100°T?
a. 110°T.
b. 284°T.
c. 104°T.
d. 090°T.

CIRCLES 25.
The automatic flight control system (AFCS) in an aircraft is coupled to the guidance outputs from an inertial navigation system (INS) and the aircraft is flying from waypoint No. 2 (60° 00'S 070° 00'W) to No. 3 (60° 00'S 080° 00'W). Comparing the initial track (° T) at 070° 00'W and the final track (° T) at 080° 00'W, the difference between them is that the initial track is approximately?

a. 9° greater than the final one.
b. 5° less than the final one.
c. 9° less than the final one.
d. 5° greater than the final one.

CIRCLES 26.
An aircraft travels from point A to point B, using the autopilot connected to the aircraft's inertial system. The coordinates of A (45° S 010°W) and B (45° S 030°W) have been entered. The true course of the aircraft on its arrival at B, to the nearest degree, is?

a. 277°.
b. 284°.
c. 263°.
d. 270°.

CIRCLES 27.
The following points are entered into an inertial navigation system (INS). WPT 1: 60° N 30°W, WPT 2: 60° N 20°W, WPT 3: 60° N 10°W. The inertial navigation system is connected to the automatic pilot on route (1-2-3). The track change when passing WPT 2 will be approximately?

a. Zero.
b. A 9° increase.
c. A 4° decrease.
d. A 9° decrease.

CIRCLES 28.
What is a rhumb line?

a. The vertex of a conformal polyformic projection.
b. The shortest distance between any two points on the Earth's surface.
c. A straight line on a Lambert's conformal chart.
d. A line on the Earth which cuts all meridians at the same angle.

CIRCLES 29.
Given: Waypoint 1. 60° S 030°W, Waypoint 2. 60° S 020°W. What will be the approximate latitude shown on the display unit of an inertial navigation system at longitude 025°W?

a. 060° 11'S.
b. 059° 49'S.
c. 060° 00'S.
d. 060° 06'S.

CIRCLES 30.
In order to fly from position A (10° 00'N, 030° 00'W) to position B (30° 00'N, 050° 00'W), maintaining a constant true course, it is necessary to fly?

a. The great-circle route.

b. A straight line plotted on a Lambert chart.
c. A rhumb line track.
d. The constant average drift route.

CIRCLES 31.
The angle between the true great-circle track and the true rhumb line track joining the following points: A (60° S 165° W) B (60° S 177° E), at the place of departure A, is?

a. 7.8°.
b. 9°.
c. 15.6°.
d. 5.2°.

CIRCLES 32.
Which of the following is true of all parallels of latitude?

a. They are neither rhumb lines nor great circles.
b. They are rhumb lines but not great circles.
c. They are both rhumb lines and great circles.
d. They are great circles but not rhumb lines.

CIRCLES 33.
An aircraft departs from position A (04° 10' S 178° 22'W) and flies northward following the meridian for 2950 NM. It then flies westward along the parallel of latitude for 382 NM to position B. The coordinates of position B are?

a. 53° 20'N 172° 38'E.
b. 45° 00'N 172° 38'E.
c. 53° 20'N 169° 22W.
d. 45° 00'N 169° 22W.

CIRCLES 34.
A Rhumb line is?

a. A line convex to the nearest pole on a Mercator projection.
b. A line on the surface of the earth cutting all meridians at the same angle.
c. The shortest distance between two points on a Polyconic projection.
d. Any straight line on a Lambert projection.

CIRCLES 35.
Given: A is N55° 000° B is N54° E010°. The average true course of the great circle is 100°. The true course of the rhumb line at point A is?

a. 096°.

b. 104°.
c. 107°.
d. 100°.

CIRCLES 36.
An aircraft passes position A (60° 00'N 120° 00'W) on route to position B (60° 00'N 140° 30'W). What is the great circle track on departure from A?

a. 279°.
b. 288°.
c. 261°.
d. 270°.

CIRCLES 37.
Parallels of latitude, except the equator, are?

a. Rhumb lines.
b. Great circles.
c. Both Rhumb lines and Great circles.
d. Neither Rhumb lines nor Great circles.

CIRCLES 38.
Given: The coordinates of the heliport at Issy les Moulineaux are N48° 50' E002° 16.5'. The coordinates of the antipodes are?

a. S41° 10' W177° 43.5'.
b. S41° 10' E177° 43.5'.
c. S48° 50' W177° 43.5'.
d. S48° 50' E177° 43.5'.

CIRCLES 39
On a Lambert Conformal Conic chart great circles that are not meridians are?

a. Curves concave to the pole of projection.
b. Straight lines within the standard parallels.
c. Curves concave to the parallel of origin.
d. Straight lines regardless of distance.

CIRCLES 40.
On a Direct Mercator chart, a rhumb line appears as a?

a. Curves convex to the Equator.
b. Curves concave to the nearest pole..
c. Straight line.
d. Complex curves cancave to the Equator..

CIRCLES 41.
On a polar stereographic projection chart showing the South Pole, a straight line joins position A (70° S 065° E) to position B (70° S 025° W). The true course on departure from position A is approximately?

a. 315°.
b. 225°.
c. 250°.
d. 135°.

CIRCLES 42.
On a Direct Mercator chart a great circle will be represented by a?

a. Straight lines.
b. Complex curves convex to the Equator.
c. Curve concave to the equator.
d. Complex curves concave to the Poles.

CIRCLES 43.
Parallels of latitude, except the equator, are?

a. Rhumb lines.
b. Complex curves.
c. Great circle.
d. Curves convex to the nearest pole.

CIRCLES 44.
On a Direct Mercator chart a great circle will be represented by a?

a. Curve convex to the equator.
b. Curve concave to the equator.
c. Complex curve.
d. Straight line.

CIRCLES 45.
Which one of the following, concerning great circles on a Direct Mercator chart, is correct?

a. With the exception of meridians and the equator, they are curves concave to the equator.
b. Their centre cannot be that of the Earth.
c. They are represented by straight lines.
d. They do not include the meridians nor the equator.

CIRCLES 46.
On a Direct Mercator chart, a rhumb line appears as a?

a. Small circle concave to the nearer pole.
b. Spiral curve.
c. Curve convex to the nearer pole.
d. Straight line.

CIRCLES 47.
Which one of the following statements is correct concerning the appearance of great circles, with the exception of meridians, on a Polar Stereographic chart whose tangency is at the pole ?

a. The higher the latitude the closer they approximate to a straight line.
b. The lower the latitude the closer they approximate to a straight line.
c. They are all represented by straight lines.
d. They can never be approximated as straight lines.

CIRCLES 48.
A perfectly straight line on a Lamberts chart is?

a. A parallel of latitude.
b. A rhumb line.
c. A meridian of longitude.
d. A great circle.

CIRCLES 49.
How do rhumb lines, (other than meridians), appear on Polar Stereographic charts?

a. As straight lines.
b. As lines concave to the nearer pole.
c. As lines convex to the nearer pole.
d. As ellipses around the pole.

CIRCLES 50.
How are great circles shown on a direct Mercator chart?

a. Curves concave to the nearest pole.
b. Curves convex to the nearest pole.
c. Straight lines.
d. Rhumb lines.

CIRCLES 51.
On a Polar Stereographic chart, the initial great circle course from A 70° N 060° W to B 70° N 060° E is approximately?

a. 030° (T).
b. 330° (T).
c. 150° (T).
d. 210° (T).

CIRCLES 52.
A straight line on a Lambert Conformal Projection chart for normal flight planning purposes?

a. Is approximately a Great Circle.
b. Is a Loxodromic line.
c. Is a Rhumb line.
d. Can only be a parallel of latitude.

CIRCLES 53.
On a Direct Mercator, rhumb lines are?

a. Ellipses.
b. Curves convex to the equator.
c. Straight lines.
d. Curves concave to the equator.

CIRCLES 54.
Which one of the following describes the appearance of rhumb lines, except meridians, on a Polar Stereographic chart?

a. Ellipses around the Pole.
b. Curves convex to the Pole.
c. Straight lines.
d. Curves concave to the Pole.

CIRCLES 55.
On a Lambert chart (standard parallels 37° N and 65° N), with respect to the straight line drawn on the map between A (N49° W030°) and B (N48° W040°), the?

a. Great circle and rhumb line are to the south.
b. Great circle and rhumb line are to the north.
c. Great circle is to the north, the rhumb line is to the south.
d. Rhumb line is to the north, the great circle is to the south.

CIRCLES 56.
On a polar stereographic projection chart showing the South Pole, a straight line joins position A (75°S 065°E) to position B (75°S 025°W). The true course on departure from position A is approximately?

a. 225°.
b. 450°.
c. 313°.
d. 320°.

CIRCLES 57.
Two positions plotted on a polar stereographic chart, A (80°N 000°) and B (70°N 102°W) are joined by a straight line whose highest latitude is reached at 035°W. At point B, the true course is?

a. 230°.
b. 203°.
c. 23°.
d. 30°.

CIRCLES 58.
An aircraft is flying with the aid of an inertial navigation system (INS) connected to the autopilot. The following two points have been entered in the INS computer: WPT 1: 60°S 030°W WPT 2: 60°S 020°W. When 025°W is passed the latitude shown on the display unit of the inertial navigation system will be?

a. 60°57'S.
b. 60°05.7'S.
c. 60°75'S.
d. 57°06'S.

CIRCLES 59
Given: Position A 45° N, ?° E, Position B 45° N, 45° 15'E Distance A-B = 350 NM, B is to the East of A, What is the longitude of position A?

a. 37° 00'E
b. 48° 00'E.
c. 51° 15'E.
d. 8° 15'E.

CIRCLES 60.
Given: Position A 45° N, ?° E, Position B 45° N, 45° 15'E Distance A-B = 200 NM, B is to the East of A, What is the longitude of position A?

a. 51° 20'E.
b. 48° 33'E.

c. 40° 32'E.
d. 51° 58'E.

CIRCLES 61.
A great circle track joins position A (57° S 140°W) and B (63° S 150°W). What is the difference between the great circle track at A and B?

a. It increases by 9°.
b. It decreases by 9°.
c. It increases by 6°.
d. It decreases by 6°.

CIRCLES 62.
Given: Position A 45° N, ?° E, Position B 45° N, 50° 15'E Distance A-B = 280 NM, B is to the East of A, What is the longitude of position A?

a. 43° 39'E.
b. 45° 57'E.
c. 52° 53'E.
d. 40° 33'E.

CIRCLES 63.
A Lambert conformal conic chart has a constant of the cone of 0.85. The initial course of a straight-line track drawn on this chart from A (40° N 050° W) to B is 043° (T) at A; course at B is 055° (T). What is the longitude of B?

a. 34°W.
b. 36°W.
c. 38°W.
d. 41°W.

CIRCLES 64.
What is the great circle track (measured from the starting position) from 70°S 035°W, to 70°S 065°E?

a. 048°T.
b. 090°T.
c. 137°T.
d. 228°T.

CIRCLES 65.
What is the longitude of a position 15 NM to the east of 58° 42'N 094° 00'W?

a. 093° 51.3'W.
b. 094° 13.0'W.

c. 093° 31.1'W.
d. 122° 31.1'W.

CIRCLES 66.
The INS position of the departure aerodrome, coordinates 45° 32.7'N 139° 46.3'W are input instead of the correct coordinates, 45° 32.7'N 139° 46.3'E. When the aircraft subsequently passes point 52° N 180°W, the longitude value shown on the INS will be?

a. 081° 26.4'W.
b. 099° 32.6'W.
c. 099° 32.6'E.
d. 081° 26.4'E.

CIRCLES 67.
An aircraft flies the following rhumb line tracks and distances from position 04° 00'N 030° 00'W: 700 NM South, then 700 NM East, then 700 NM North, then 700 NM West. The final position of the aircraft is?

a. 04° 00'N 030° 30'W.
b. 03° 58'N 030° 02'W.
c. 04° 00'N 029° 55'W.
d. 04° 00'N 029° 55'E.

CIRCLES 68.
Which of the following differences in latitude will produce the greatest difference between the initial great circle track and the mean great circle track, between two points, which are separated by 15° change of longitude?

a. 30°S and 25°S.
b. 30°S and 30°N.
c. 60°N and 55°N.
d. 60°N and 60°S.

CIRCLES 69.
What is the final position after the following rhumb line tracks and distances have been followed from position 60° 00'N 030° 00'W? South for 360 NM, East for 360 NM, North for 360 NM, West for 360 NM?

a. 59° 00'N 090° 00'W.
b. 60° 00'N 031° 48'W.
c. 60° 00'N 031° 48'E.
d. 59° 00'N 060° 00'W.

CIRCLES 70.
The Great Circle bearing of 'B' (70° S 050° E), from 'A' (70° S 030° W), is approximately?

a. 150° (T).
b. 090° (T).
c. 115° (T).
d. 128° (T).

CIRCLES 71.
An aircraft at position 60° N 005° W tracks 090° (T) for 350 km. On completion of the flight the longitude will be?

a. 002° 10'W.
b. 000° 15'E.
c. 001° 19'E.
d. 001° 19'W.

CIRCLES 72.
An aircraft autopilot is coupled to the INS steering from Waypoint 1 at 60°N 050°W. to Waypoint 2 at 60°N 020°W. What is the approximate latitude of the aircraft on passing 35°W?

a. 60°51'N.
b. 60°05'N.
c. 60°30'N.
d. 59°15'N.

CIRCLES 73.
A Lambert conformal conic chart has a constant of the cone of 0.80. A straight line course drawn on this chart from A (53° N 004°W) to B is 075° at A Course at B is 095° (T). What is the longitude of B?

a. 021° E.
b. 009° 36'E.
c. 021° W.
d. 019° E.

CIRCLES 74.
An aircraft is flying with the aid of an inertial navigation system (INS) connected to the autopilot. The following two points have been entered in the INS computer: WPT 1: 60° N 040°W, WPT 2: 60° N 010°W. When 025° W has been passed, the latitude shown on the display unit of the inertial navigation system will be?

a. 60° 55.0'N.

b. 59° 49.0'N.
c. 60° 11.0'N.
d. 60° 05.0'N.

CIRCLES 75.
The 'departure' between positions 60°N 100° E and 60° N 'x' is 1200 NM. What is the longitude of 'x'?

a. 140°E.
b. 140°W.
c. 175° W.
d. 175° E.

CIRCLES 76.
Given: The autopilot is coupled to the Inertial Navigation System steering from Waypoint 1 at 60°N 040°W, to Waypoint 2 at 60°N 030°W, to Waypoint 3 at 60°N 020°W. What is the approximate track change on passing WPT 2?

a. Zero.
b. 4° decrease.
c. 9° decrease.
d. 9° increase.

CIRCLES 77.
An aircraft autopilot is coupled to the INS steering from Waypoint 1 at 60°N 060°W, to Waypoint 2 at 60°N 030°W. What is the approximate latitude of the aircraft on passing 45°W?

a. 60°54'N.
b. 59°15'N.
c. 60°23'N.
d. 60°19'N.

CIRCLES 78.
The 'departure' between positions 60°N 90° E and 60° N 'x' is 1200 NM. What is the longitude of 'x'?

a. 130°E.
b. 130°W.
c. 145° E.
d. 145° W.

CIRCLES 79.
The 'departure' between positions 60°N 160° E and 60° N 'x' is 1800 NM East. What is the longitude of 'x'?

a. 120° E.
b. 140° E.
c. 140°W.
d. 120°W.

CIRCLES 80.
An aircraft with its autopilot connected to its inertial system, flies from point A at 50°S 010°W, to point B at 50°S 030°W. What is the true course of the aircraft (to the nearest degree) when it arrives at B?

a. 265°.
b. 275°.
c. 279°.
d. 286°.

CIRCLES 81.
The automatic flight control system (AFCS) in an aircraft is coupled to the guidance outputs from an inertial navigation system (INS). The aircraft is flying between inserted waypoints No. 3 (65° 00'N 020° 00'W) and No.4 (65° 00'N 030° 00'W). With DSRTK/STS selected on the CDU, to the nearest whole degree, the initial track read-out from waypoint No. 3 will be?

a. 266°.
b. 270°.
c. 275°.
d. 279°.

CIRCLES 82.
An aeroplane flies from A (59° S 140° W) to B (61° S 149° W) with a TAS of 480 kt. The autopilot is engaged and coupled with an Inertial Navigation System in which AB track is active. On route AB, the true track?

a. Varies by 10°.
b. Decreases by 6°.
c. Varies by 4°.
d. Increases by 8°.

CIRCLES 83.
The automatic flight control system is coupled to the guidance outputs from an inertial navigation system. Which pair of latitudes will give the greatest difference between initial track read-out and the average true course given, in each case, a difference of longitude of 10°?

a. 35° S to 35°N.
b. 30° S to 25°S.
c. 60° N to 60°N.

d. 65° N to 65°N.

CIRCLES 84.
Position A is 58°N 030°W and position B is 51°N 020°W. What is the rhumb line bearing from A to B, if the great circle track from A to B, measured at A, is 100°T?

a. 110°T.
b. 284°T.
c. 104°T.
d. 090°T.

CIRCLES 85.
The automatic flight control system (AFCS) in an aircraft is coupled to the guidance outputs from an inertial navigation system (INS) and the aircraft is flying from waypoint No. 2 (70° 00'S 070° 00'W) to No. 3 (70° 00'S 080° 00'W). Comparing the initial track (° T) at 070° 00'W and the final track (° T) at 080° 00'W, the difference between them is that the initial track is approximately?

a. 9.4° greater than the final one.
b. 5.4° less than the final one.
c. 9.4° less than the final one.
d. 5.4° greater than the final one.

CIRCLES 86.
An aircraft travels from point A to point B, using the autopilot connected to the aircraft's inertial system. The coordinates of A (65° S 010°W) and B (65° S 020°W) have been entered. The true course of the aircraft on its arrival at B, to the nearest degree, is?

a. 275°.
b. 286°.
c. 265°.
d. 270°.

CIRCLES 87.
The following points are entered into an inertial navigation system (INS). WPT 1: 70° N 50°W, WPT 2: 70° N 40°W, WPT 3: 70° N 30°W. The inertial navigation system is connected to the automatic pilot on route (1-2-3). The track change when passing WPT 2 will be approximately?

a. Zero.
b. A 9° increase.
c. A 6° decrease.
d. A 9° decrease.

CIRCLES 88.
What is a rhumb line?

a. A line on the Earth which cuts all meridians at the same angle.
b. The vertex of a conformal polyformic projection.
c. The shortest distance between any two points on the Earth's surface.
d. A straight line on a Lambert's conformal chart.

CIRCLES 89.
Given: Waypoint 1. 75° S 030°W, Waypoint 2. 75° S 020°W. What will be the approximate latitude shown on the display unit of an inertial navigation system at longitude 025°W?

a. 060° 11'S.
b. 059° 49'S.
c. 060° 00'S.
d. 060° 07'S.

CIRCLES 90.
In order to fly from position A (10° 00'N, 030° 00'W) to position B (30° 00'N, 050° 00'W), maintaining a constant true course, it is necessary to fly?

a. A rhumb line track.
b. The constant average drift route.
c. The great-circle route.
d. A straight line plotted on a Lambert chart.

CIRCLES 91.
The angle between the true great-circle track and the true rhumb line track joining the following points: A (70° S 160° W) B (70° S 179° E), at the place of departure A, is?

a. 8.9°.
b. 6.9°.
c. 15.6°.
d. 5.2°.

CIRCLES 92.
Which of the following is true of all parallels of latitude?

a. They are both rhumb lines and great circles.
b. They are great circles but not rhumb lines.
c. They are neither rhumb lines nor great circles.
d. They are rhumb lines but not great circles.

CIRCLES 93.
An aircraft departs from position A (09° 10' S 178° 22'W) and flies northward following the meridian for 2750 NM. It then flies westward along the parallel of latitude for 400 NM to position B. The coordinates of position B are?

a. 53° 20'N 172° 38'E.
b. 36° 40'N 173° 19'E.
c. 36° 20'N 169° 22W.
d. 40° 00'N 169° 22W.

CIRCLES 94.
A Rhumb line is?

a. The shortest distance between two points on a Polyconic projection.
b. Any straight line on a Lambert projection.
c. A line convex to the nearest pole on a Mercator projection.
d. A line on the surface of the earth cutting all meridians at the same angle.

CIRCLES 95.
Given: A is N55° 000° B is N54° E010°. The average true course of the great circle is 100°. The true course of the rhumb line at point A is?

a. 100°.
b. 096°.
c. 104°.
d. 107°.

CIRCLES 96.
An aircraft passes position A (75° 00'N 120° 00'W) on route to position B (75° 00'N 140° 30'W). What is the great circle track on departure from A?

a. 280°.
b. 285°.
c. 267°.
d. 275°.

CIRCLES 97.
Parallels of latitude, except the equator, are?

a. Both Rhumb lines and Great circles.
b. Neither Rhumb lines nor Great circles.
c. Rhumb lines.
d. Great circles.

CIRCLES 98.
Given: The coordinates a position are N48° 50' E002° 16.5'. The coordinates of the antipodes of that position are?

a. S48° 50' W177° 43.5'.
b. S48° 50' E177° 43.5'.
c. S41° 10' W177° 43.5'.
d. S41° 10' E177° 43.5'.

CIRCLES 99
On a Lambert Conformal Conic chart great circles that are not meridians are?

a. Curves concave to the parallel of origin.
b. Straight lines regardless of distance.
c. Curves concave to the pole of projection.
d. Straight lines within the standard parallels.

CIRCLES 100.
On a Direct Mercator chart, a rhumb line appears as a?

a. Curves convex to the Equator.
b. Complex curves concave to the Equator.
c. Curves concave to the nearest pole.
d. Straight line.

CIRCLES 101.
On a polar stereographic projection chart showing the South Pole, a straight line joins position A (75° S 065° E) to position B (75° S 025° W). The true course on departure from position A is approximately?

a. 315°.
b. 225°.
c. 250°.
d. 135°.

CIRCLES 102.
On a Direct Mercator chart a great circle will be represented by a?

a. Curve concave to the equator.
b. Complex curves concave to the Poles.
c. Straight lines.
d. Complex curves convex to the Equator.

CIRCLES 103.
Parallels of latitude, except the equator, are?

a. Great circle.
b. Curves convex to the nearest pole.
c. Rhumb lines.
d. Complex curves.

CIRCLES 104.
On a Direct Mercator chart a great circle will be represented by a?

a. Complex curve.
b. Straight line.
c. Curve convex to the equator.
d. Curve concave to the equator.

CIRCLES 105.
Which one of the following, concerning great circles on a Direct Mercator chart, is correct?

a. They are represented by straight lines.
b. They do not include the meridians nor the equator.
c. With the exception of meridians and the equator, they are curves concave to the equator.
d. Their centre cannot be that of the Earth.

CIRCLES 106.
On a Direct Mercator chart, a rhumb line appears as a?

a. Curve convex to the nearer pole.
b. Straight line.
c. Small circle concave to the nearer pole.
d. Spiral curve.

CIRCLES 107.
Which one of the following statements is correct concerning the appearance of great circles, with the exception of meridians, on a Polar Stereographic chart whose tangency is at the pole ?

a. They are all represented by straight lines.
b. They can never be approximated as straight lines.
c. The higher the latitude the closer they approximate to a straight line.
d. The lower the latitude the closer they approximate to a straight line.

CIRCLES 108.
A perfectly straight line on a Lamberts chart is?

a. A meridian of longitude.
b. A great circle.
c. A parallel of latitude.
d. A rhumb line.

CIRCLES 109.
How do rhumb lines, other than meridians appear on Polar Stereo charts?

a. As lines convex to the nearer pole.
b. As straight lines.
c. As lines concave to the nearer pole.
d. As ellipses around the pole.

CIRCLES 110.
How are great circles shown on a direct Mercator chart?

a. Curves concave to the nearest pole.
b. Rhumb lines.
c. Curves convex to the nearest pole.
d. Straight lines.

CIRCLES 111.
On a Polar Stereographic chart, the initial great circle course from A 75° N 060° W to B 75° N 060° E is approximately?

a. 032° (T).
b. 330° (T).
c. 150° (T).
d. 210° (T).

CIRCLES 112.
A straight line on a Lambert Conformal Projection chart for normal flight planning purposes?

a. Is a Rhumb line.
b. Can only be a parallel of latitude.
c. Is approximately a Great Circle.
d. Is a Loxodromic line.

CIRCLES 113.
On a Direct Mercator, rhumb lines are?

a. Ellipses.
b. Curves concave to the equator.
c. Curves convex to the equator.

d. Straight lines.

CIRCLES 114.
Which one of the following describes the appearance of rhumb lines, except meridians, on a Polar Stereographic chart?

a. Ellipses around the Pole.
b. Curves concave to the Pole.
c. Curves convex to the Pole.
d. Straight lines.

CIRCLES 115.
On a Lambert chart (standard parallels 37° N and 65° N), with respect to the straight line drawn on the map between A (N49° W030°) and B (N48° W040°), the?

a. Great circle and rhumb line are to the north.
b. Great circle is to the north, the rhumb line is to the south.
c. Rhumb line is to the north, the great circle is to the south.
d. Great circle and rhumb line are to the south.

CIRCLES 116.
On a polar stereographic projection chart showing the South Pole, a straight line joins position A (75°S 045°E) to position B (75°S 025°W). The true course on departure from position A is approximately?

a. 235°.
b. 450°.
c. 320°.
d. 225°.

CIRCLES 117.
Two positions plotted on a polar stereographic chart, A (80°N 000°) and B (70°N 102°W) are joined by a straight line whose highest latitude is reached at 035°W. At point B, the true course is?

a. 230°.
b. 203°.
c. 23°.
d. 30°.

CIRCLES 118.
An aircraft is flying with the aid of an inertial navigation system (INS) connected to the autopilot. The following two points have been entered in the INS

computer: WPT 1: 60°S 030°W WPT 2: 60°S 020°W. When 025°W is passed the latitude shown on the display unit of the inertial navigation system will be?

a. 60°57'S.
b. 60°05.7'S.
c. 60°75'S.
d. 57°06'S.

CIRCLES 119
Given: Position A 45° N, ?° E, Position B 45° N, 45° 15'E Distance A-B = 350 NM, B is to the East of A, What is the longitude of position A?

a. 37° 00'E.
b. 48° 00'E.
c. 51° 15'E.
d. 8° 15'E.

CIRCLES 120.
Given: Position A 45° N, ?° E, Position B 45° N, 45° 15'E Distance A-B = 200 NM, B is to the East of A, What is the longitude of position A?

a. 51° 20'E.
b. 48° 33'E.
c. 38° 32'E.
d. 51° 58'E.

MAGNETISM & COMPASSES 1.
When accelerating on an easterly heading in the Northern hemisphere, the compass card of a direct reading magnetic compass will turn?

a. Anti-clockwise giving an apparent turn toward the south.
b. Clockwise giving an apparent turn toward the north.
c. Clockwise giving an apparent turn toward the south.
d. Anti-clockwise giving an apparent turn toward the north.

MAGNETISM & COMPASSES 2.
Sensitivity of a direct reading magnetic compass is?

a. Proportional to the vertical component of the magnetic field of the earth.
b. Inversely proportional to the vertical and horizontal components of the magnetic field of the earth.
c. Inversely proportional to horizontal component of the magnetic field of the earth.
d. Proportional to horizontal component of the magnetic field of the earth.

MAGNETISM & COMPASSES 3.
When turning right from 320° (C) to 050° (C) in the Southern hemisphere, the reading of a direct reading magnetic compass will?

a. Over-indicate the turn and liquid swirl will decrease the effect.
b. Under-indicate the turn and liquid swirl will decrease the effect.
c. Over-indicate the turn and liquid swirl will increase the effect.
d. Under-indicate the turn and liquid swirl will increase the effect.

MAGNETISM & COMPASSES 4.
An aircraft is completing a timed turn in the Northern hemisphere from 330°C to 040°C. After the turn has been stopped but before the direct indicating magnetic compass settles down, will the compass (1) over-read or under-read, and (2) is this error increased or decreased by liquid swirl?

a. (1) over-read (2) increased.
b. (1) over-read (2) decreased.
c. (1) under-read (2) increased.
d. (1) under-read (2) decreased.

MAGNETISM & COMPASSES 5.
What is magnetic variation?

a. The angle between Magnetic North and True North.
b. The angle between Magnetic Heading and Magnetic North.
c. The angle between the direction indicated on the compass and Magnetic North.
d. The angle between True North and Compass North.

MAGNETISM & COMPASSES 6.
An aircraft in the northern hemisphere makes an accurate rate one turn to the right/starboard. If the initial heading was 330°, after 30 seconds of the turn the direct reading magnetic compass should read?

a. 060°.
b. More than 060°.
c. More or less than 060° depending on the pendulous suspension used.
d. Less than 060°.

MAGNETISM & COMPASSES 7.
At the magnetic equator?

a. Deviation is zero.
b. The isogonal is an agonic line.
c. Dip is zero.
d. Variation is zero.

MAGNETISM & COMPASSES 8.
Which of the following is the correct conversion from True to Compass?

	T	V	M	D	C
a.	130	2W	128	-1	127.
b.	130	2E	128	-1	129.
c.	130	2E	132	-1	133.
d.	130	2W	132	-1	133.

MAGNETISM & COMPASSES 9.
The horizontal component of the earth's magnetic field?

a. Weakens with increasing distance from the magnetic poles.
b. Weakens with increasing distance from the nearer magnetic pole.
c. Is approximately the same at all magnetic latitudes less than 60°.
d. Is approximately the same at magnetic latitudes 50° N and 50°S.

MAGNETISM & COMPASSES 10.
What is the maximum value of dip that can be caused by the vertical component of terrestrial magnetism?

a. 45°.
b. 60°.
c. 90°.
d. 180°.

MAGNETISM & COMPASSES 11.
Which of the following is correct when the variation is West?

a. True North is East of Magnetic North.
b. Magnetic North is West of Compass North.
c. True North is West of Magnetic North.
d. Compass North is West of Magnetic North.

MAGNETISM & COMPASSES 12.
A line drawn on a chart which joins all points where the value of magnetic variation is zero is called an?

a. Agonic line.
b. Aclinic line.
c. Isogonal.
d. Isotach.

MAGNETISM & COMPASSES 13.
How is the direct reading magnetic compass made aperiodic or dead beat?

a. Using a pendulous suspension system for the magnetic assembly.
b. Using the lowest possible viscosity liquid in the compass.
c. Using short magnets.
d. Positioning the magnetic assembly mass close to the compass point and using damping wires.

MAGNETISM & COMPASSES 14.
Isogonals converge at the?

a. Magnetic equator.
b. North magnetic pole only.
c. North and South magnetic poles only.
d. North and South geographic and magnetic poles.

MAGNETISM & COMPASSES 15.
Why does the value of magnetic variation on a chart change with time?

a. Movements of the magnetic poles, cause it to increase.
b. Movements of the magnetic poles, cause it to increase or decrease.
c. Increases in the magnetic field, cause it to increase.
d. Reductions in the magnetic field, cause it to decrease.

MAGNETISM & COMPASSES 16.
What is the compass heading if true track is 348°, drift is 17° left, variation is 32° W, and deviation is 4°E?

a. 009°.
b. 033°.
c. 339°.
d. 337°.

MAGNETISM & COMPASSES 17.
The value of magnetic variation?

a. Must be 0° at the magnetic equator.
b. Varies between a maximum of 45° East and 45° West.
c. Cannot exceed 90°.
d. Has a maximum of 180°.

MAGNETISM & COMPASSES 18.
At what point on the earth is a magnetic compass most effective?

a. At the geographic equator.
b. Close to the magnetic North Pole.

c. Close to the magnetic South Pole.
d. Approximately midway between the magnetic poles.

MAGNETISM & COMPASSES 19.
Which of the following is true of the value of magnetic variation?

a. It is 0° at the magnetic equator.
b. It varies between 45° East and 45° West.
c. It cannot exceed 90°.
d. It cannot exceed 180°.

MAGNETISM & COMPASSES 20.
An aircraft in the northern hemisphere is making an accurate rate one turn to the right. If the initial heading was 135°, after 30 seconds the direct reading magnetic compass should read?

a. 225°.
b. Less than 225°.
c. More or less than 225° depending on the pendulous suspension used.
d. More than 225°.

MAGNETISM & COMPASSES 21.
When using a direct reading magnetic compass in the northern hemisphere?

a. A longitudinal acceleration on a Westerly heading causes an apparent turn to the South.
b. A longitudinal acceleration on an Easterly heading causes an apparent turn to the North.
c. A longitudinal deceleration on a Westerly heading causes an apparent turn to the North.
d. A longitudinal acceleration on an Easterly heading causes an apparent turn to the South.

MAGNETISM & COMPASSES 22.
What is the main cause of permanent magnetism in aircraft?

a. Electrical currents in internal wiring and electrical storms.
b. Exposure to the terrestrial magnetic field during normal operation.
c. The combined effect of aircraft electrical equipment and the terrestrial magnetic field.
d. Hammering, and the effect of the earth's magnetic field, whilst under construction.

MAGNETISM & COMPASSES 23.
When accelerating on a westerly heading in the northern hemisphere, the compass card of a direct reading magnetic compass will turn?

a. Clockwise giving an apparent turn towards the north.
b. Clockwise giving an apparent turn towards the south.
c. Anti-clockwise giving an apparent turn towards the north.
d. Anti-clockwise giving an apparent turn towards the south.

MAGNETISM & COMPASSES 24.
What is the main advantage of a remote indicating compass compared to a direct reading compass is that it?

a. It has less moving parts.
b. It requires less maintenance.
c. It magnifies the earth's magnetic field in order to attain greater accuracy.
d. It senses, rather than seeks, the magnetic meridian.

MAGNETISM & COMPASSES 25.
Where do the isogonal lines converge?

a. At the both the North and South Geographic Poles.
b. At both the North and South Magnetic Poles.
c. Only at the Magnetic equator.
d. Only at the Magnetic North Pole.

MAGNETISM & COMPASSES 26.
When decelerating on a westerly heading in the Northern hemisphere, the compass card of a direct reading magnetic compass will turn?

a. Anti-clockwise giving an apparent turn towards the north.
b. Clockwise giving an apparent turn toward the south.
c. Anti-clockwise giving an apparent turn towards the south.
d. Clockwise giving an apparent turn towards the north.

MAGNETISM & COMPASSES 27.
Why are the detector units in slaved gyro compasses usually fitted in or close to the wingtips of an aircraft?

a. In order to isolate the detector unit from the terrestrial magnetic field.
b. In order to reduce or eliminate turning and acceleration errors.
c. Having one detector in each wingtip, cancels out compass deviation.
d. In order to isolate the detector unit from the aircraft deviation sources.

MAGNETISM & COMPASSES 28.
Which of the following statements is true of the Earth's magnetic field?

a. It may be permanent, temporary, transient.
b. It does not affect aircraft deviation.
c. It acts like a large blue magnetic pole in Northern Canada.
d. The dip angle is between the vertical and the total magnetic force.

MAGNETISM & COMPASSES 29.
In a remote indicating compass system the amount of deviation caused by aircraft magnetism and electrical circuits may be minimised by?

a. Positioning the master unit in the centre of the aircraft.
b. Using a vertically mounted gyroscope.
c. Mounting the detector unit in the wingtip.
d. The use of repeater cards.

MAGNETISM & COMPASSES 30.
What is the angle between True North and Magnetic North?

a. Drift.
b. Variation.
c. Dip.
d. Deviation.

MAGNETISM & COMPASSES 31.
An aircraft using a direct reading magnetic compass in the Northern hemisphere starts a Rate 1 turn to the right from 145°C. What will be the indicated heading if it rolls out of the turn after 30 seconds?

a. It is not possible to answer this question using the information given.
b. Greater than 235°.
c. Less than 235°.
d. 235°.

MAGNETISM & COMPASSES 32.
The sensitivity of a direct reading compass varies?

a. Inversely with the vertical component of the earth's magnetic field.
b. Directly with the horizontal component of the earth's magnetic field.
c. Directly with the vertical component of the earth's magnetic field.
d. Inversely with both vertical and horizontal components of the earth's magnetic field

MAGNETISM & COMPASSES 33.
An aircraft's compass must be swung?

a. If the aircraft has been in the hangar for a long time and has been moved several times.
b. If the aircraft has been subjected to hammering.
c. Every maintenance inspection.
d. After a change of theatre of operations which leads to a change of magnetic longitude.

MAGNETISM & COMPASSES 34.
What is the dip angle at the South Magnetic Pole?

a. 00°.
b. 90°.
c. 180°.
d. 64°.

MAGNETISM & COMPASSES 35.
At the magnetic equator, when accelerating after take off on heading West, a direct reading compass?

a. Underreads.
b. The heading indicates a turn to the south.
c. Indicates the correct heading.
d. Overreads the heading.

MAGNETISM & COMPASSES 36.
What is the name for a line of equal magnetic variation?

a. An isocline.
b. An isogonal.
c. An isogriv.
d. An isovar.

MAGNETISM & COMPASSES 37.
The agonic line?

a. Is midway between the magnetic North and South Poles.
b. Follows the geographic equator.
c. Is the shorter distance between the respective True and Magnetic North and South Poles.
d. Follows separate paths out of the North polar regions, one currently running through Western Europe and the other through the USA.

MAGNETISM & COMPASSES 38.
Isogrives are lines that connect positions that have?

a. The same variation.
b. O° magnetic dip.
c. The same grivation.
d. The same horizontal magnetic field strength.

MAGNETISM & COMPASSES 39.
Which of the following statements concerning earth magnetism is completely correct?

a. An isogonal is a line which connects places of equal dip; the aclinic is the line of zero magnetic dip.
b. An isogonal is a line which connects places with the same magnetic variation; the aclinic connects places with the same magnetic field strength.
c. An isogonal is a line which connects places with the same magnetic variation; the aclinic is the line of zero magnetic dip.
d. An isogonal is a line which connects places with the same magnetic variation; the agonic line is the line of zero magnetic dip.

MAGNETISM & COMPASSES 40.
An aircraft is accelerating in the Northern hemisphere heading West. Will a direct reading magnetic compass (1) over-read or under-read and (2) Indicate a turn to the north or to the south?

a. (1) over-read (2) north.
b. (1) over-read (2) south.
c. (1) under-read (2) north.
d. (1) under-read (2) south.

MAGNETISM & COMPASSES 41.
What is the advantage of the remote indicating compass (slaved gyro compass) over the direct reading magnetic compass?

a. It is lighter.
b. It is connected to a source of electrical power and so is more accurate.
c. It senses the earth's magnetic field rather than seeks it, so is more sensitive.
d. It is not affected by aircraft deviation.

MAGNETISM & COMPASSES 42.
Isogrivs on a chart indicate lines of?

a. Zero magnetic variation.

b. Equal magnetic dip.
c. Equal horizontal directive force.
d. Equal grivation.

MAGNETISM & COMPASSES 43.
Compass deviation is defined as the angle between?

a. True North and Compass North.
b. The horizontal and the total intensity of the earth's magnetic field.
c. Magnetic North and Compass North.
d. True North and Magnetic North.

MAGNETISM & COMPASSES 44.
Isogonic lines connect positions that have?

a. The same angle of magnetic dip.
b. The same variation.
c. 0° variation.
d. The same elevation.

MAGNETISM & COMPASSES 45.
When an aircraft on a westerly heading on the northern hemisphere accelerates, the effect of the acceleration error causes the magnetic compass to?

a. Lag behind the turning rate of the aircraft.
b. Indicate a turn towards the south.
c. Turn faster than the actual turning rate of the aircraft.
d. Indicate a turn towards the north.

MAGNETISM & COMPASSES 46.
When is the magnetic compass most effective?

a. In the region of the magnetic North Pole.
b. On the geographic equator.
c. About midway between the magnetic poles.
d. In the region of the magnetic South Pole.

MAGNETISM & COMPASSES 47.
An Agonic line is a line that connects?

a. Positions that have 0° variation.
b. Positions that have the same variation.
c. Points of equal magnetic dip.
d. Points of equal magnetic horizontal field strength.

MAGNETISM & COMPASSES 48.
A direct reading compass should be swung when?

a. The aircraft has made more than a stated number of landings.
b. There is a large, and permanent, change in magnetic latitude.
c. There is a large change in magnetic longitude.
d. The aircraft is stored for a long period and is frequently moved.

MAGNETISM & COMPASSES 49.
The annunciator of a remote indicating compass system is used when?

a. Synchronising the magnetic and gyro compass elements.
b. Compensating for deviation.
c. Setting local magnetic variation.
d. Setting the 'heading' pointer.

MAGNETISM & COMPASSES 50.
The direct reading magnetic compass is made aperiodic (dead beat) by?

a. Using long magnets.
b. Pendulous suspension of the magnetic assembly.
c. Keeping the magnetic assembly mass close to the compass point and by using damping wires.
d. Using the lowest acceptable viscosity compass liquid.

MAGNETISM & COMPASSES 51.
The Earth can be considered as being a magnet with the?

a. Red pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface.
b. Blue pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface.
c. Red pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface.
d. Blue pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface.

MAGNETISM & COMPASSES 52.
Deviation applied to magnetic heading gives?

a. Magnetic track.
b. Compass heading.
c. True heading.
d. Magnetic course.

MAGNETISM & COMPASSES 53.
The angle between Magnetic North and Compass North is called?

a. Compass deviation.
b. Compass error.
c. Magnetic variation.
d. Alignment error.

MAGNETISM & COMPASSES 54.
Which of the following is an occasion for carrying out a compass swing on a Direct Reading Compass?

a. Before an aircraft goes on any flight that involves a large change of magnetic latitude.
b. After any of the aircraft radio equipment has been changed due to unserviceability.
c. Whenever an aircraft carries a large freight load regardless of its content.
d. After an aircraft has passed through a severe electrical storm, or has been struck by lightning.

MAGNETISM & COMPASSES 55.
Which of the following statements is correct concerning the effect of turning errors on a direct reading compass?

a. Turning errors are greatest on east/west headings, and are greatest at high latitudes.
b. Turning errors are greatest on north/south headings, and are greatest at high latitudes.
c. Turning errors are greatest on east/west headings, and are least at high latitudes.
d. Turning errors are greatest on north/south headings, and are least at high latitudes.

MAGNETISM & COMPASSES 56.
Which one of the following is an advantage of a remote reading compass as compared with a standby compass?

a. It eliminates the effect of turning and acceleration errors by pendulously suspending the detector unit.
b. It is more reliable because it is operated electrically and power is always available from sources within the aircraft.
c. It senses the magnetic meridian instead of seeking it, increasing compass sensitivity.
d. It is lighter than a direct reading compass because it employs, apart from the detector unit, existing aircraft equipment.

MAGNETISM & COMPASSES 57.
Complete the following statement regarding magnetic variation. The charted values of magnetic variation on earth normally change annually due to?

a. Magnetic pole movement causing numerical values at all locations to increase or decrease.
b. Magnetic pole movement causing numerical values at all locations to increase.
c. A reducing field strength causing numerical values at all locations to decrease.
d. An increasing field strength causing numerical values at all locations to increase.

MAGNETISM & COMPASSES 58.
The north and south magnetic poles are the only positions on the earth's surface where?

a. The value of magnetic variation equals 90°.
b. A freely suspended compass needle will stand vertical.
c. Isogonals converge.
d. A freely suspended compass needle will stand horizontal.

MAGNETISM & COMPASSES 59.
The main reason for mounting the detector unit of a remote reading compass in the wingtip of an aeroplane is?

a. To ensure that the unit is in the most accessible position on the aircraft for ease of maintenance.
b. By having detector units on both wingtips, to cancel out the deviation effects caused by the aircraft strucure.
c. To minimise the amount of deviation caused by aircraft magnetism and electrical circuits.
d. To maximise the units exposure to the earth's magnetic field.

MAGNETISM & COMPASSES 60.
The main advantage of a remote indicating compass over a direct reading compass is that it?

a. Requires less maintenance.
b. Is able to magnify the earth's magnetic field in order to attain greater accuracy.
c. Senses, rather than seeks, the magnetic meridian.
d. Has less moving parts.

MAGNETISM & COMPASSES 61.
The main reason for usually mounting the detector unit of a remote indicating compass in the wingtip of an aeroplane is to?

a. Place it where it will not be subjected to electrical or magnetic interference from the aircraft.
b. Reduce the amount of deviation caused by aircraft magnetism and electrical circuits.
c. Facilitate easy maintenance of the unit and increase its exposure to the Earth's magnetic field.
d. Place it in a position where there is no electrical wiring to cause deviation errors.

MAGNETISM & COMPASSES 62.
Permanent magnetism in aircraft arises chiefly from?

a. The combined effect of aircraft electrical equipment and the earth's magnetic field.
b. The effect of internal wiring and exposure to electrical storms.
c. Hammering, and the effect of the earth's magnetic field, whilst under construction.
d. Exposure to the earth's magnetic field during normal operation.

MAGNETISM & COMPASSES 63.
Which of the following statements concerning the earth's magnetic field is completely correct?

a. Dip is the angle between total magnetic field and vertical field component.
b. The blue pole of the earth's magnetic field is situated in North Canada.
c. At the earth's magnetic equator, the inclination varies depending on whether the geographic equator is north or south of the magnetic equator.
d. The earth's magnetic field can be classified as transient, semi-permanent or permanent.

MAGNETISM & COMPASSES 64.
The lines on the earth's surface that join points of equal magnetic variation are called?

a. Isogrives.
b. Isoclines.
c. Isogonals.
d. Isotachs.

MAGNETISM & COMPASSES 65.
Concerning direct reading magnetic compasses, in the northern hemisphere, it can be said that?

a. On an Easterly heading, a longitudinal acceleration causes an apparent turn to the South.
b. On a Westerly heading, a longitudinal acceleration causes an apparent turn to the South.
c. On a Westerly heading, a longitudinal deceleration causes an apparent turn to the North.
d. On an Easterly heading, a longitudinal acceleration causes an apparent turn to the North.

MAGNETISM & COMPASSES 66.
At a specific location, the value of magnetic variation?

a. Depends on the type of compass installed.
b. Depends on the magnetic heading.
c. Depends on the true heading.
d. Varies slowly over time.

MAGNETISM & COMPASSES 67.
Isogonals are lines of equal?

a. Wind velocity.
b. Magnetic variation.
c. Compass deviation.
d. Pressure.

MAGNETISM & COMPASSES 68.
In northern hemisphere, during an acceleration in an easterly direction, the magnetic compass will indicate?

a. A heading of East.
b. A decrease in heading.
c. An increase in heading.
d. An apparent turn to the South.

MAGNETISM & COMPASSES 69.
A negative (westerly) magnetic variation signifies that?

a. Compass North is East of Magnetic North.
b. Compass North is West of Magnetic North.
c. True North is East of Magnetic North.
d. True North is West of Magnetic North.

MAGNETISM & COMPASSES 70.
The purpose of compass check swing is to?

a. Cancel out the horizontal component of the earth's magnetic field.
b. Cancel out the effects of the magnetic fields found on board the aeroplane.
c. Measure the angle between Magnetic North and Compass North.
d. Cancel out the vertical component of the earth's magnetic field.

MAGNETISM & COMPASSES 71.
You are turning from 330°C to 040°C in the northern hemisphere. At the time that you stop the turn, but before the compass settles down, will the compass over-read or under-read (i), and will liquid swirl tend to increase or decrease the error (ii)?

a. (i) Over-read (ii) Decrease.
b. (i) Over-read (ii) Increase.
c. (i) Under-read (ii) Decrease.
d. (i) Under-read (ii) Increase.

MAGNETISM & COMPASSES 72.
The sensitivity of a direct reading magnetic compass is?

a. Inversely proportional to the horizontal and vertical components of the earth's magnetic field.
b. Directly proportional to the horizontal and vertical components of the earth's magnetic field.
c. Proportional to the horizontal component of the earth's magnetic field.
d. Proportional to the vertical component of the earth's magnetic field.

MAGNETISM & COMPASSES 73.
The magnitude of variation?

a. Has a maximum of 90°.
b. Has a maximum of 180°.
c. Is zero at the equator.
d. Has a maximum value of 45°E or 45°W.

MAGNETISM & COMPASSES 74.
Which of the following statements about the earth's magnetic field is most accurate?

a. It has no effect on aircraft deviation.
b. It may be permanent, temporary or transient.
c. It acts as if there were a large red magnetic pole in northern Canada.
d. The dip angle is the angle between the total magnetic field and the earth horizontal.

MAGNETISM & COMPASSES 75.
At the magnetic equator?

a. Variation is zero.
b. Deviation is zero.
c. Dip is zero.
d. The agonic line is also an isogonal line.

MAGNETISM & COMPASSES 76.
Which of the following is the best definition of magnetic variation?

a. The angle between the magnetic heading and the magnetic north.
b. The angle between the magnetic north and the true north.
c. The angle between the true north and the compass north.
d. The angle between the direction indicated on a magnetic compass and magnetic north.

MAGNETISM & COMPASSES 77.
At which point on the surface of the earth is a magnetic compass most effective?

a. On the geographical equator.
b. Close to the magnetic north pole.
c. Close to the magnetic South Pole.
d. Approximately midway between the magnetic poles.

MAGNETISM & COMPASSES 78.
Which of the following statements about the earth's magnetic field is most accurate?

a. It has no effect on aircraft deviation.
b. It may be permanent, temporary or transient.
c. It acts as if there were a large blue magnetic pole in northern Canada.
d. The dip angle is the angle between the total magnetic field and the earth vertical.

MAGNETISM & COMPASSES 79.
An aircraft is accelerating on a westerly heading in the northern hemisphere. Will the compass be over-reading or under-reading (i) and will it indicate a turn to the north or south (ii)?

a. (i) Over-reading (ii) South turn.
b. (i) Over-reading (ii) North turn.
c. (i) Under-reading (ii) South turn.
d. (i) Under-reading (ii) North turn.

MAGNETISM & COMPASSES 80.
What advantage does the remote indicating or slaved gyro compass have in comparison to the direct reading magnetic compass?

a. Not affected by deviation.
b. Lighter.
c. More accurate because it is not connected to any source of electrical power.
d. Is more sensitive because it senses the earth's magnetic field rather than seeking it.

MAGNETISM & COMPASSES 81.
An aircraft is on a heading of 135°C using a direct reading magnetic compass in the northern hemisphere. After carrying out a rate 1 turn for 30 seconds on what indicated heading should it roll out of the turn?

a. 225.
b. More than 225.
c. Less than 225.
d. It is not possible to answer this question using the data provided.

MAGNETISM & COMPASSES 82.
What are indicated by isogrivs on a chart?

a. Lines of equal grivation.
b. Lines of zero magnetic variation.
c. Lines of equal magnetic dip.
d. Lines of equal effective horizontal magnetic force.

MAGNETISM & COMPASSES 83.
The variation is 20°E and you are heading 345°M when you take a radar bearing of an island which is 30° left of the nose. What bearing do you plot?

a. 155°T.
b. 160°T.
c. 170°T.
d. 180°T.

MAGNETISM & COMPASSES 84.
The agonic line?

a. Follows the geographical equator.
b. Follows two separate paths out of the North polar regions. One currently passes through the USA while the other passes through Western Europe.
c. Is midway between the magnetic north and South Poles.

d. Is the shorter of the distance between the true and Magnetic North and South Poles.

MAGNETISM & COMPASSES 85.
What is the angle between True North and Magnetic North known as?

a. Alignment error.
b. Dip.
c. Deviation
d. Variation.

MAGNETISM & COMPASSES 86.
Why does the value of magnetic variation on a chart changes with time?

a. Reduction in the magnetic field, causing a decrease.
b. Movement of the magnetic poles, which can cause either an increase or a decrease.
c. Movement of the magnetic poles, causing an increase.
a. Increase in the magnetic field, causing an increase.

MAGNETISM & COMPASSES 87.
Where do isogonal lines converge?

a. At the North and South Magnetic Poles.
b. At the Magnetic equator.
c. A the North Magnetic Pole.
d. At the North and South Magnetic and Geographical Poles.

MAGNETISM & COMPASSES 88.
Why are the detector units of a slaved gyro-compass system usually located in or near to the aircraft wingtips?

a. Having one detector unit in each wingtip, cancels out compass deviations.
b. To isolate the detector unit from the aircraft deviation sources.
c. To isolate the detector unit from the magnetic field of the Earth.
d. To minimise turning and acceleration errors.

MAGNETISM & COMPASSES 89.
What is the maximum possible value of Dip Angle?

a. 45°.
b. 66°.
c. 90°.
d. 180°.